Programming Fundamental

Home Tasks - Week 06

Task 01:

Write a C++ program that **takes n numbers from the user, stores them in an array, and passes the array to a function that calculates and prints the sum of all elements.**

Test Cases:

```
PS E:\TA ship\Week 6\Problems> cd "e:\TA ship
ask1.cpp -o task1 } ; if ($?) { .\task1 }
Enter number of elements: 4
Enter 4 numbers:
12
12
43
1
Sum of all elements: 68


PS E:\TA ship\Week 6\Problems> cd "e:\TA ship\We
ask1.cpp -o task1 } ; if ($?) { .\task1 }
Enter number of elements: 6
Enter 6 numbers:
3
2
4
5
7
98
Sum of all elements: 119       cd "e:\TA ship\We


ask1.cpp -o task1 } ; if ($?) { .\task1 }
Enter number of elements: 2
Enter 2 numbers:
4
6
Sum of all elements: 10
```

Task 02:

Write a C++ program that **takes n numbers from the user, stores them in an array, and passes the array to a function that counts and prints how many numbers are even.**

Test Cases:

```
PS E:\TA ship\Week 6\Problems> cd "e:\TA
ask2.cpp -o Task2 } ; if ($?) { .\Task2
Enter number of elements: 2
Enter 2 numbers:
2
3
Total Even Numbers: 1

PS E:\TA ship\Week 6\Problems> cd "e:\TA
ask2.cpp -o Task2 } ; if ($?) { .\Task2 }
Enter number of elements: 6
Enter 6 numbers:
23
12
4
13
56
2
Total Even Numbers: 4          cd "e:\TA

ask2.cpp -o Task2 } ; if ($?) { .\Task2 }
Enter number of elements: 8
Enter 8 numbers:
34
34
2
24
65
5
7
3
Total Even Numbers: 4
```

Task 03:

Write a C++ program that **takes n numbers from the user, stores them in an array, and passes the array to a function that finds and prints the largest and Smallest number in the array**

Test Cases:

```
PS E:\TA ship\Week 6\Problems> cd "e:\
ask3.cpp -o task3 } ; if ($?) { .\task
Enter number of elements: 3
Enter 3 numbers:
5
43
4
Largest Number: 43
Smallest Number: 4
```

```
PS E:\TA ship\Week 6\Problems> cd "e:
ask3.cpp -o task3 } ; if ($?) { .\tas
Enter number of elements: 6
Enter 6 numbers:
24
456
13
678987
34
1
Largest Number: 678987
Smallest Number: 1
```

```
PS E:\TA ship\Week 6\Problems> cd "e:\TA ship\
ask3.cpp -o task3 } ; if ($?) { .\task3 }
Enter number of elements: 8
Enter 8 numbers:
2
4
4
23
4
5
6
0
Largest Number: 23
Smallest Number: 0
```

Task 04:

Write a C++ program that **takes the names of 5 students from the user and stores them in a string array. The program should then display all student names.**

Test Cases:

```
PS E:\TA ship\Week 6\Problems> cd "e:\TA ship\Week 6\
ask4.cpp -o task4 } ; if ($?) { .\task4 }
Enter names of 5 students:
ali
hamza
hamdan
waqar
essa
Student Names are:
ali
hamza
hamdan
waqar
essa
PS E:\TA ship\Week 6\Problems>
```

Task 05:

Write a C++ program that takes the names of n customers from the user and stores them in a string array. Ask the user to enter a letter and count how many names start with that letter.

```
ask5.cpp -o task5 } ; if ($?) { .\task5 }
Enter number of customers: 4
Enter the names of 4 customers:
hamza
essa
haram
waqar
Enter a letter to check: h
Total names starting with 'h': 2
PS E:\TA ship\Week 6\Problems>
```

```
PS E:\TA ship\Week 6\Problems> cd "e:\TA sl
ask5.cpp -o task5 } ; if ($?) { .\task5 }
Enter number of customers: 3
Enter the names of 3 customers:
ali
hamza
essa
Enter a letter to check: n
Total names starting with 'n': 0
```

Task 06:

Write a C++ program that **takes the names of n students from the user and stores them in a string array. The program should then display the names in alphabetical order.**

```
PS E:\TA ship\Week 6\Problems> cd "e:
ask6.cpp -o task6 } ; if ($?) { .\tas
Enter number of students: 3
Enter names of 3 students:
ali
essa
dawood
Students in alphabetical order:
ali
dawood
essa
```

```
PS E:\TA ship\Week 6\Problems> cd "e:\TA ship\
ask6.cpp -o task6 } ; if ($?) { .\task6 }
Enter number of students: 5
Enter names of 5 students:
Haram
Essa
Azyan
Waqar
awais
Students in alphabetical order:
Azyan
Essa
Haram
Waqar
awais
```

Task 07:

A shop has 4 products.

- Arrays: productNames[], prices[], quantity[]
- Task: Print product name, price, and stock quantity. Also, calculate total value per product (price * quantity).

Example:

Input: productNames = ["Apple", "Banana", "Orange", "Mango"]
prices = [0.5, 0.3, 0.8, 1.2]
quantity = [50, 100, 30, 20]

Output: Apple: $0.5, 50 in stock, Total value: $25
Banana: $0.3, 100 in stock, Total value: $30

Test Cases:

```
Enter number of products: 2

Enter name of product 1: Apple
Enter price of Apple: $23
Enter quantity of Apple: 12

Enter name of product 2: pineapple
Enter price of pineapple: $12
Enter quantity of pineapple: 1

Product Inventory Report
------------------------
Apple: $23.00, 12 in stock, Total value: $276.00
pineapple: $12.00, 1 in stock, Total value: $12.00
```

```
ask7.cpp -o task7 } ; if ($?) { .\task7 }
Enter number of products: 3

Enter name of product 1: Butter
Enter price of Butter: $12
Enter quantity of Butter: 4

Enter name of product 2: Egg
Enter price of Egg: $1
Enter quantity of Egg: 3

Enter name of product 3: Bread
Enter price of Bread: $24
Enter quantity of Bread: 2

Product Inventory Report
------------------------
Butter: $12.00, 4 in stock, Total value: $48.00
Egg: $1.00, 3 in stock, Total value: $3.00
Bread: $24.00, 2 in stock, Total value: $48.00
```

Task 08:

Airline wants to track flights.

- Arrays: flightNumber[], destination[], seatsAvailable[]

- Task: Print flight info and find flights with seats less than 5.

```
ask8.cpp -o task8 } ; if ($?) { .\task8 }
Enter number of flights: 2

Enter flight number for flight 1: 1344
Enter destination for flight 1344: lahore
Enter seats available for flight 1344: 3

Enter flight number for flight 2: 13
Enter destination for flight 13: Karachi
Enter seats available for flight 13: 1

Flight Information:
-------------------
Flight 1344 to lahore has 3 seats available.
Flight 13 to Karachi has 1 seats available.

Flights with less than 5 seats available:
-----------------------------------------
Flight 1344 to lahore has only 3 seats left!
Flight 13 to Karachi has only 1 seats left!
```

```
ask8.cpp -o task8 } ; if ($?) { .\task8 }
Enter number of flights: 1

Enter flight number for flight 1: 23
Enter destination for flight 23: Moscow
Enter seats available for flight 23: 2345

Flight Information:
-------------------
Flight 23 to Moscow has 2345 seats available.

Flights with less than 5 seats available:
-----------------------------------------
No flights with less than 5 seats available.
```

Task 09:

Write a C++ Program To stimulate a Library System using a Menu-Driven Approach.
The program should display the following menu:

1.Add Books
2.View Books
3.Borrow Books
4.View Books

5.Exit

Use Array to Store book: